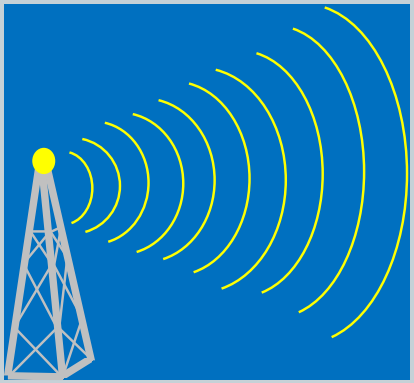
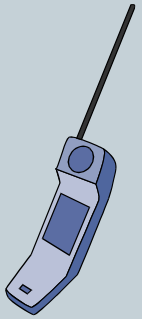
COULOMB'S LAW AND ELECTRIC FIELD INTENSITY

Class Continued by:
Prof. Dr. Md. Jahedul Islam
Cell: 01792451681
Email: jahed@cuet.ac.bd

Introduction to Electromagnetic Fields





• transmitter and receiver are connected by a “field.”

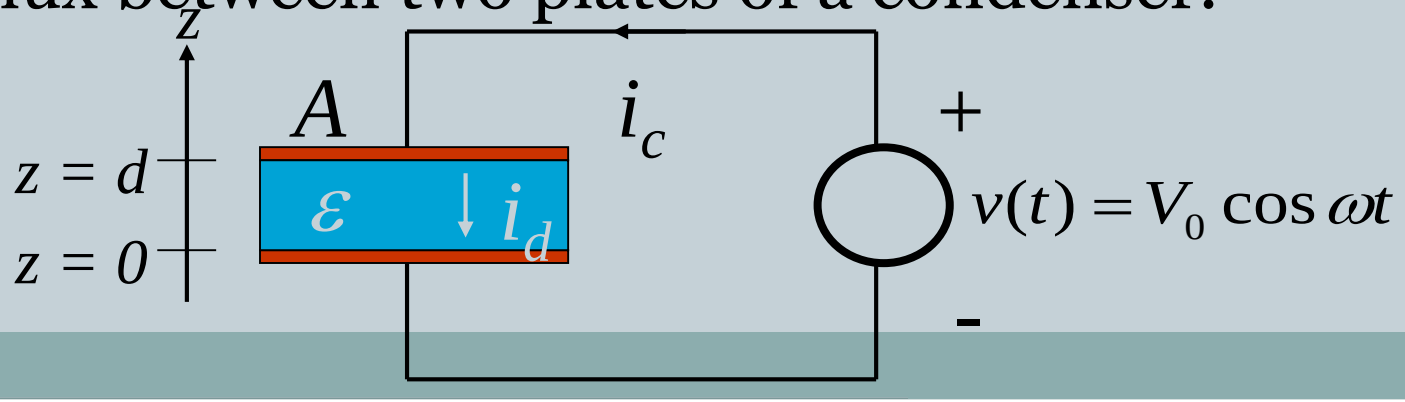
Fundamental Terms

- A ***field*** is a spatial distribution of a quantity; in general, it can be either *scalar* or *vector* in nature.

- Electric and magnetic fields:
  - Are vector fields with three spatial components.
  - Vary as a function of position in 3D space as well as time.
  - Are governed by partial differential equations derived from Maxwell's equations.

Fundamental Terms

- Conduction Current: When a potential is applied in a conductor the charges move and cause a current flow. This entire flow of current is called conduction current.
- Displacement current: The displacement current is defined as $i_d = C\frac{dv}{dt} = i_c$; rate of change of electric flux between two plates of a condenser.



Fundamental Terms

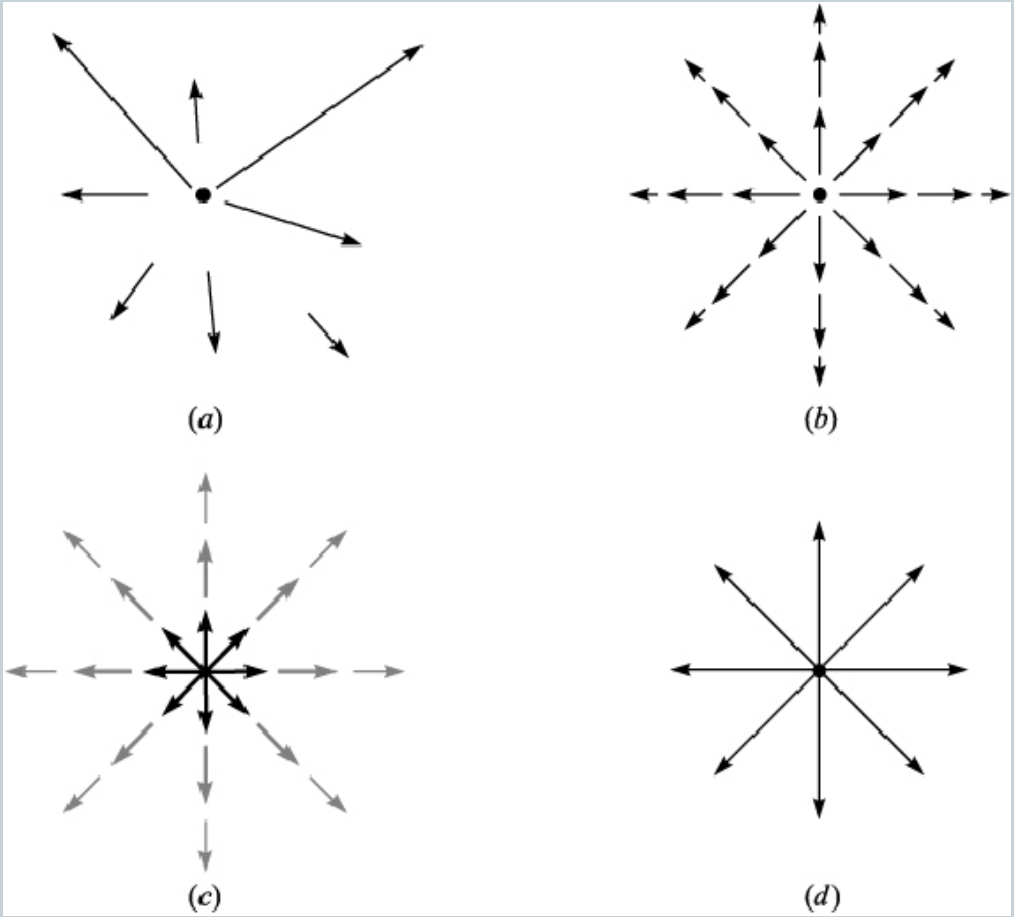
- Electric Field: A region around a charged particle or object within which a force would be exerted on other charged particles or objects.

- Electric Flux: In common language flux refers to the flow or stream of any thing from one point to another point. In the similar way electric flux is the total number of lines of force passing through a surface.

STREAMLINES AND SKETCHES OF FIELDS

- (a) One very poor sketch,
- (b) and (c) two fair sketches, and
- (d) the usual form of streamline sketch.

In the last form, the arrows show the direction of the field at every point along the line, and the spacing of the lines is inversely proportional to the strength of the field.



Electrical Flux

- The streamlines show the direction of the force on a test charge at every point
- It is convenient to assume these lines a physical significance and thinking of them as flux lines.
- An electric flux which streams away symmetrically from a point charge and coincide with the streamlines
- To visualize wherever an electric field is present we can use this flux.

ELECTRIC FLUX DENSITY

- Electric flux density described as "lines per square meter and measured in coulombs per square meter ," (for each line is due to one coulomb)
- It is denoted by the letter D, which was originally chosen because of the alternate names of displacement flux density or displacement density.
- The electric flux density D is a vector field
- The direction of D at a point is the direction of the flux lines at that point, and the magnitude is given by the number of flux lines crossing a surface normal to the lines divided by the surface area.

Fundamental Electromagnetic Laws/Equations

- Coulomb's Law
- Ampere's Circuital law
- Biot-Savart Law
- Laplace and Poisson's equation
- Maxwell's equation
- Stoke's theorem (optional)

COULOMB'S LAW

- The force between two very small objects separated in a vacuum or free space by a distance which is large compared to their size is proportional to the charge on each and inversely proportional to the square of the distance between them

$$F = K\frac{Q_1 Q_2}{R^2}$$

COULOMB'S LAW (contd.)

- If the International System of Units (SI) is used, Q is measured in coulombs (C), R is in meters (m), and the force should be newtons (N). This will be achieved if the constant of proportionality k is written as

$$K = \frac{1}{4\pi\varepsilon_0}$$

$$\varepsilon_0 = 8.854 \times 10^{-12} = \frac{1}{36\pi} 10^{-9} F/m$$

COULOMB'S LAW (contd.)

- Coulomb's law is now

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}$$

Example: Two charges of Q1 = 3 nC and Q2 = 10 nC have a distance of 5μm in vacuum. We desire the force exerted on Q2 by Q1.

Electric Field Intensity

- The vector force on a unit positive charge is called ELECTRIC FIELD INTENSITY
  - Consider one charge fixed in position, say Q_1
  - Move a test charge, Q_t slowly around Q_1
  - This test charge is displaying the existence of a force field. The force on it is given by Coulomb's law,

$$F = \frac{Q_1 Q_t}{4\pi\varepsilon_0 R_{12}^2} a_{12}$$

Electric Field Intensity (Contd.)

- Writing this force as a force per unit charge gives

$$\frac{\mathbf{F}_t}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 R_{1t}^2}\mathbf{a}_{1t}$$

The quantity on the right side is a function only of Q1 and the directed line segment from Q1 to the position of the test charge. This describes a vector field and is called the electric field intensity.

Potential difference

- We define potential difference, V as the work done (by an external source) in moving a unit positive charge from one point to another in an electric field

$$\text{Potential difference} = V = -\int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

- It is measured in joules per coulomb, for which the volt is defined as a more common unit, abbreviated as V.

Potential difference (Cont..)

- The potential difference between points A and B is

$$V_{AB} = -\int_{B}^{A} \mathbf{E} \cdot d\mathbf{L} \quad \text{V}$$

- V_{AB} signifies the potential difference between points A and B and is the work done in moving the unit charge from B (last named) to A (first named).
- V_{AB} is positive if work is done in carrying the positive charge from B to A.

Potential difference (Cont..)

- From the line-charge example of the last section we found that the work done in taking a charge Q from ρ = a to ρ = b along a radial path

$$W = \frac{Q\rho_L}{2\pi\epsilon_0}\ln\frac{b}{a}$$

- Thus, the potential difference between points at ρ = a to ρ = b is

$$V_{ab} = \frac{W}{Q} = \frac{\rho_L}{2\pi\epsilon_0}\ln\frac{b}{a}$$

Conservative property of electrostatic field

- If a charge is taken around any closed path, no net work is done. Mathematically this may be written as the closed line integral

$$\oint E.dL = 0$$

- This is the conservative property of electrostatic field.

Electric Dipole

- Two closely spaced point charges of equal magnitude and opposite sign, is called an electric dipole

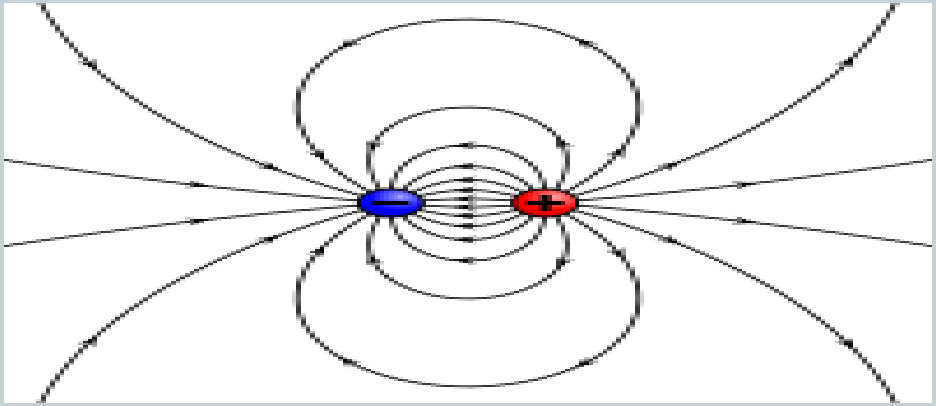


Fig: Electric dipole field of two point-charges with small distance

Poisson's and Laplace's equations

Obtaining Poisson's equation is exceedingly simple, for from the point form of Gauss's law,

$$\nabla \cdot \mathbf{D} = \rho_v \tag{1}$$

the definition of D,

$$\mathbf{D} = \epsilon \mathbf{E} \tag{2}$$

and the gradient relationship,

$$\mathbf{E} = -\nabla V \tag{3}$$

by substitution we have

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon \mathbf{E}) = -\nabla \cdot (\epsilon \nabla V) = \rho_v$$

Poisson's and Laplace's equations (cont..)

or

$$\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon} \tag{4}$$

for a homogeneous region in which ϵ is constant.

Equation (4) is *Poisson's equation*, but the "double ∇" operation must be interpreted and expanded, at least in cartesian coordinates, before the equation can be useful. In cartesian coordinates,

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla V = \frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Poisson's and Laplace's equations (cont..)

and therefore

$$\nabla \cdot \nabla V = \frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial V}{\partial z}\right)$$

$$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \qquad (5)$$

Usually the operation $\nabla \cdot \nabla$ is abbreviated ∇^2 (and pronounced "del squared"), a good reminder of the second-order partial derivatives appearing in (5), and we have

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon} \qquad (6)$$

in cartesian coordinates.

Equation (6) is generally named as Poisson's equation

Poisson's and Laplace's equations (cont..)

If $\rho_v = 0$, indicating zero *volume* charge density, but allowing point charges, line charge, and surface charge density to exist at singular locations as sources of the field, then

$$\nabla^2 V = 0 \tag{7}$$

which is *Laplace's equation*. The ∇^2 operation is called the *Laplacian of V*.

In cartesian coordinates Laplace's equation is

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \qquad \text{(cartesian)} \tag{8}$$

The Laplacian in different co-ordinate system

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\left(\frac{\partial^2 V}{\partial \phi^2}\right) + \frac{\partial^2 V}{\partial z^2} \qquad \text{(cylindrical)}$$

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2} \qquad \text{(spherical)}$$